

# NATHANIEL JOSEPH PALAIA

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## EDUCATION

**The Ohio State University** | B.S. Aerospace Engineering with Honors | GPA: 3.98

Aug 2024 – May 2028

*Minor: Robotics and Autonomous Systems*

## SKILLS

**GNSS & RF:** GPS/GNSS Data Processing, NMEA/UBX Protocol, ExpressLRS 2.4 GHz, Serial/UART, MarvelMind Positioning.

**Systems & Software:** Embedded Systems, Electronics, Sensor Fusion (Optical Flow + GNSS), UAV Autopilot Config, Python (GEE, GeoPandas, Matplotlib), C/C++, Google Earth Engine, GIS, MATLAB, Simulink, Google Colab, VS Code, NI Multisim (Circuit Design), Code/Device Troubleshooting, Systems Level Troubleshooting.

**CAD & Manufacturing:** PCBA Design, Prototyping, Autodesk Inventor & AutoCAD (CAD/CAM), Onshape, 3D Printing, soldering experience, G-Code.

**Soft Skills:** Technical writing, Scientific Communication, Data Organization, Team Collaboration, Documentation, Independent Learning.

## RESEARCH EXPERIENCE

**Research Assistant – Autonomous UAS Fleet Development** | *UAS Lab, OSU MAE*

Feb 2025 – Present

- Established and now lead an indoor GPS-denied UAS research program. Educate 6 undergraduate engineers on UAS development.
- Coordinated 10" and 5" prop quadcopter assembly, autopilot configuration (Arducopter), avionics and subsystems integration, ultrasonic beacon and flight controller link.
- Implemented lowpass filtering capabilities in the power distribution circuit to prevent electronic failures due to voltage spikes.
- Successfully troubleshooted and diagnosed electrical failure in an ESC through systematic analysis of soldering joints and motor tests.
- Contributed to integrating GPS-denied indoor positioning using MarvelMind ultrasonic beacons, for position data with  $\pm 2$  cm accuracy.
- Developed NMEA serial pipeline between ultrasonic beacon grid and UAV-mounted beacons to emulate GPS indoors; established ExpressLRS 2.4 GHz RF for remote control flight, configured optical flow sensor fusion for position hold, and ensured electronic systems compatibility using Ardupilot Extended Kalman Filtering parameters.
- Designing an UAV enhanced with dual RGB and IR simultaneous image capturing and onboard AI computing capabilities for a Ph.D. project involving remote sensing of prescribed burns.
- Earned FAA Part 107 Commercial Remote Pilot Certificate.

**Research Assistant – Remote Sensing & Geospatial Analysis** | *Geospatial Hydrology Lab, OSU FABE*

Apr 2025 – Present

- Developed an original geospatial data product to proxy forest regeneration after fire ability by combining 5 environmental data layers (lithology, slope, aspect, vegetation, and soil) into a combined layer.
- Processed multi-temporal satellite imagery (large datasets) using Python (GEE, GeoPandas, Matplotlib); computed spectral indices (EVI, NDVI) to validate an original geospatial product for post-wildfire forest recovery prediction derived from geospatial data using ArcGIS and saw trends of EVI increasing with higher values of the data product (suggests accurate prediction of regeneration ability).
- Presented research at the 3rd Annual Interdisciplinary Water Research Symposium (OSU) and the AGU 2025 Annual Meeting (New Orleans, LA); currently drafting a formal research paper.

## ENGINEERING PROJECTS

**Autonomous Robotic Design Project** | *OSU First Year Engineering Honors Program*

Jan 2025 – Apr 2025

- Designed, 3D-modeled, and integrated a fully autonomous robot to complete a garden-themed obstacle course; contributed to motorized/electronic subsystem (servos, motors, CdS cell sensor) integration.
- Produced a comprehensive design report covering all prototypes, milestones, electrical schematics, and competition results.

**Video Game Design Project (C++)** | *OSU First Year Engineering Honors Program*

Nov 2024 – Dec 2024

- Created the GUI for a small bug-themed video game using C++ coding techniques.
- Contributed to object-oriented programming to add features (such as clickable objects) to the game.

**Anti-Forgetfulness Pen Case** | *SkillsUSA Student Lead Innovation Competition*

Jan 2024 – May 2024

- Designed and assembled a PCBA controlling an LED triggered when the pen was not in place and the case was closed. The circuit utilized integrated circuit components to handle digital logic.
- 3D modeled and printed enclosure in Autodesk Inventor CAD.
- Drafted a full patent following industry standards, including technical drawings produced in Autodesk Inventor.

## LEADERSHIP

**Deputy Director, Research Issues Committee** | *OSU Undergraduate Student Government*

Sep 2024 – Oct 2025

- Led initiatives to expand undergraduate research access campus-wide; co-organized a Research Forum connecting students with 5 faculty researchers actively seeking undergrad assistants.

**Student Leader** | *Cru at Ohio State*

Aug 2024 – Present

- Mentor undergraduate students in personal and academic development; conduct weekly one-on-one meetings to support peers.

**Sigma Gamma Tau** | *Honors Aerospace Society, Ohio State Chapter*

Aug 2024 – Present

- Engage in professional development opportunities, develop comradery within aerospace engineering at Ohio State University.

**Eagle Scout Project** | *Scouting America*

Aug 2019 – Oct 2019

- Led \$1,500+ project in designing and building raised flower beds and concrete pavement for a local retirement home community.